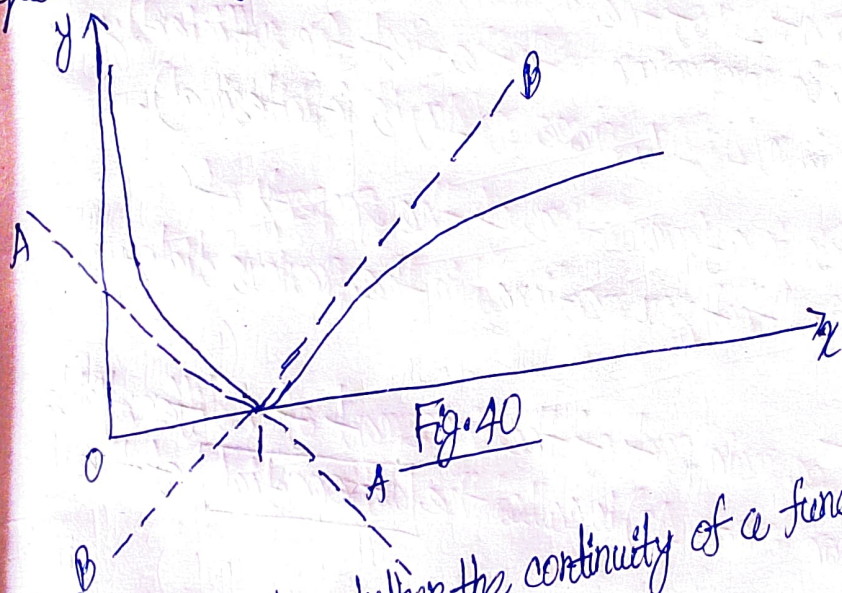


must be continuous at  $x$ . The following theorem could be proved rigorously.

**Theorem:** The continuity of a function  $f(x)$  at a point  $x$  is a necessary condition for the existence of  $f'(x)$  at  $x$ .

However, we shall not give the proof of this theorem here. The simple qualitative arguments given above will suffice.



**Reader:** I wonder whether the continuity of a function is also a sufficient condition for its differentiability.

**Author:** No, it is not. Consider, for example, the function  $y = |\log x|$ . It is sufficient only to look at its graph (Fig. 40) to conclude that at  $x=1$  the tangent to the graph of the function is, strictly speaking, nonexistent (on approaching  $x=1$  from the left we have one tangent, viz, the straight line AA, while on approaching  $x=1$  from the right we have another tangent, viz, the straight line BB). It means that  $y = |\log x|$  does not have a derivative at  $x=1$ , although the function is continuous at this point.

In conclusion, let us turn to one interesting property of differentiable functions. Let  $f(x)$  be a differentiable function, and its increment  $\Delta f$  at  $x$  be related to the increment  $\Delta x$  of the argument as follows:

$$\Delta f = f'(x) \Delta x + \eta(\Delta x) \Delta x$$

(6) where  $\eta(\Delta x)$  is a function of  $\Delta x$ . By dividing both parts of (6) by  $\Delta x$ , we obtain